Architectures Differentiate — and Where Their Premises Fail

Status Report III — the confirmatory local cell at $N=5$, a protocol-loss counterfactual, and swarm vote anatomy

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Abstract

The first confirmatory cell of the pre-registered architecture study is complete: the frozen frontier tier (v2.1) × three architectures × $N=5$ on the local environment — 540 runs, every cell exactly 5/5, one config hash. Two validation results anchor the report. First, the tier froze at monolithic domain aggregates of 42/58/67% and *reproduced* at 43/58/67% — the calibration methodology holds on the production serving stack. Second, that stack mattered: a pre-sweep check caught the v2 code domain drifting from 48% (LM Studio) to 35% (Ollama), forcing a mechanical re-source (v2.1, 58%) and a new guardrail — *calibration bands are verified on the serving stack that produces the study data*. The headline: swarm 61.1% > monolithic 56.1% > agentic 47.2%, at 3.0× and 1.9× monolithic's token cost respectively. Both multi-agent results, however, decompose into mechanism findings rather than architecture wins. Agentic's deficit is partly *protocol-compliance loss*: the reasoning model buries its APPROVE verdict mid-prose, the framework's first-line parser misses it, and the resulting "false revision" loops destroy correct, already-approved answers. A pre-declared counterfactual re-parse of saved traces recovers agentic math from 33% to 42% — parity with monolithic — while code and multihop are untouched: the residual gap is real architecture cost. Swarm's win dissolves under vote anatomy: across all 180 swarm rows the vote is never genuinely contested (three 2–1 code splits from serving jitter aside); its multihop edge comes from two tasks flipped deterministically by the peer prompt. At temperature 0 the pre-registered swarm is a system-prompt ablation at triple cost, not an ensemble — motivating a proposed SWARM 2.0 arm (peers at $T>0$), its amendment drafted for logging before any of its data exist. Scope has shifted: the confirmatory environments are Apple Metal (M5 Max, M4) and university HPC (CUDA), whose sweep is running as this report is written; the Azure cell is reframed from a GPU VM to an *exploratory managed-inference arm* (Azure AI Foundry, serverless DeepSeek-R1) to run in parallel with SWARM 2.0.

1 Context and changes since Report II

Design, hypotheses, and metrics remain frozen in `PRE_REGISTRATION.md` (2026-07-01); every deviation is amendment-logged. Since Report II (2026-07-09):

Serving stack pinned to Ollama; tier re-frozen as v2.1. All remaining environments serve via native Ollama (CUDA boxes and HPC have no LM Studio path). A production-stack calibration check — monolithic $N=5$ over the frozen v2 code domain — measured **35% on Ollama vs. 48% in the original LM Studio calibration**. The drift is real (same 4 items solid, 7 floored) and out-of-band, so the domain re-sourced mechanically under the domain-aggregate rule: v2.1 code is a deterministic 6+6 mix of first-validating BigCodeBench-Hard and BigCodeBench items, which calibrated to **58% in band on Ollama** (7 items solved, 5 unsolved). Math and multihop carry unchanged. Methodological consequence, now a guardrail: difficulty is relative not only to the *model* (Report II) but to the *serving stack*; bands are verified on the production stack before any multi-architecture sweep.

Environment scope reshaped. The pre-registered Azure cell is reframed (amendment pending advisor confirmation): a rented GPU VM would duplicate the HPC cell's serving stack on worse hardware, so the confirmatory environments are Apple Metal (M5 Max primary; M4 Mini as a cross-hardware point) and university HPC (NVIDIA A40, CUDA). Azure re-enters as an *exploratory* managed-inference arm instead: Azure AI Foundry's serverless full DeepSeek-R1 (the 14B distill is not serverless-deployable), which cannot hold the pinned artifact — precision, batching, and decoding are not controllable — and is therefore framed as "self-hosted pinned artifact vs. managed serving stack," measuring accuracy deltas, latency distribution, determinism at temperature 0, and \$/token (verified pricing \$1.35/\$5.40 per M tokens; a full $N=5$ sweep estimates ∼\$27). A Shadow cloud-PC (RTX A4500) served as the CUDA rehearsal environment ($N=1$, 2026-07-12), validating NVML telemetry and surfacing two harness fixes (the CUDA telemetry dependency install; the reasoning-model `thinking`-field answer recovery) before the confirmatory runs.

2 Confirmatory local cell: 540 runs

Frozen v2.1 tier (36 items) × 3 architectures × $N=5$ on the M5 Max (48 GB, Metal, Ollama). Integrity: all 108 cells exactly 5/5 trials, single config hash, zero missing rows. Failure mix: 211 reasoning errors, 23 budget exhaustions, 10 code-harness (tool) errors.

The calibration reproduced. Monolithic domain aggregates landed at 43/58/67% against calibration's 42/58/67%

Table 1: Local confirmatory cell (N=5): accuracy by domain, and mean cost.

	Math	Code	Multihop	**Overall**
Monolithic	**43%**	**58%**	67%	56.1%
Agentic	33%	50%	58%	47.2%
Swarm	42%	**58%**	**83%**	**61.1%**

Mean output tokens/run: 2755 / 5148 / 8177 (1× / 1.9× / 3.0×)
Median latency/run: 48 s / 88 s / 140 s at ∼43 tok/s (Metal)

— a one-point deviation in one domain across 180 fresh runs. The tier holds its band placement on the production stack; the pre-declared selection rule measured what it claimed to measure.

3 Mechanism finding 1: the false revision

Agentic trails everywhere — but instrumentation shows part of the deficit is not the *architecture* at all. The pinned verifier prompt demands the verdict on the first line ("APPROVE or REVISE"); the backend parses it with `startswith("APPROVE")`. DeepSeek-R1 explains first and verdicts later, so verbal approvals routinely fail the parse; worse, any verifier turn that exhausts the token budget returns chain-of-thought (the empty-content fallback), which *structurally* cannot begin with APPROVE. Each "false revision" forces another Executor→Verifier loop against an already-approved answer; the loop frequently ends with an empty final answer. In the cleanest exhibit the executor answered 70 (gold: 70), the verifier wrote ". . . is indeed 70. APPROVE", and all five trials scored zero.

Per a pre-declared amendment (logged before any counterfactual number was computed), a mechanical re-parse of the saved traces asks what each agentic row would have scored under lenient verdict reading (last standalone APPROVE/REVISE token decides; the approved candidate is graded by the pinned grader; primary rows untouched).

Table 2: Agentic: pinned vs. counterfactual (local N=5).

Domain	Pinned	Counterfactual	Δ
Math	33%	42%	+8.3 pts
Code	50%	50%	0
Multihop	58%	58%	0
Overall	47.2%	50.0%	+2.8 pts

Five rows flip, all math, all wrong→right; none flip down. The decomposition is sharp: on math, protocol loss explains agentic's *entire* deficit (counterfactual 42% = parity with monolithic 43% and swarm 42%); on code and multihop the gap is genuine architecture cost — verifier judgment errors and revision-loop budget burn. The general lesson: agentic is the only architecture whose *control flow* runs through model prose, and reasoning models are unreliable at strict output protocols; frameworks that key control flow on exact tokens silently convert approvals into destructive revision loops.

4 Mechanism finding 2: swarm never votes

Report II's exploratory data hinted the swarm's majority vote might be inert at temperature 0; the confirmatory cell settles it. Across all 180 swarm rows: no genuinely contested votes (60/60 math and 60/60 multihop unanimous or single-key; three 2–1 code splits from serving jitter; 28 math abstention events from correlated budget exhaustion — all three peers truncating together in 8 of the 10 affected rows). Greedy decoding ignores the per-peer seed offsets, so the peers are near-clones and the pre-registered "independent samples" premise fails mechanically.

The headline multihop win (83% vs. 67%) reduces to exactly two tasks, each flipped 0/5→5/5 deterministically by the peer *prompt*: one where the peer prompt elicits a different reasoning path (monolithic answers the wrong hop), and one of answer specificity (monolithic's under-specified league name fails the string grader; peers emit the full name). At temperature 0 the pre-registered swarm is a system-prompt ablation at 3× cost, not an ensemble.

Consequence: a **SWARM 2.0** arm is proposed, its amendment drafted for logging *before any of its data exist* — identical topology, prompts, and tasks, but peers decode at $T>0$ with per-peer seeds (temperature chosen by a small pre-declared probe), restoring the premise the majority vote requires. Swarm-1.0 rows remain the valid measurement of the pre-registered design; the paper reports them as a diagnosed negative result with the vote-anatomy evidence above.

5 Cross-hardware reproducibility

Three environments have now run the frontier tier at the pinned 14B:

Table 3: Cross-environment view (v2.1 except Shadow = v2; N as shown).

Env (accel.)	N	Mono	Agentic	Swarm
M5 Max (Metal)	5	56%	47%	61%
M4 Mini (Metal)	1	58%	—	61%
Shadow A4500 (CUDA, v2)	1	44%	39%	50%

Aggregates reproduce across accelerators and boxes (the Shadow cell ran the earlier v2 code domain, depressing its absolute numbers); per-item results do not reproduce at N=1 — temperature 0 does not force identical outputs across Metal and CUDA. Decode throughput is memory-bandwidth arithmetic: the base M4 (∼120 GB/s) holds ∼10.4 tok/s on the 9 GB Q4 model, the M5 Max ∼43 tok/s, the A4500 (∼640 GB/s) ∼39 tok/s — each near the ceiling its bandwidth predicts.

6 Secondary metric: LLM-as-judge

The different-family judge (Llama-3.2-3B) processed all 540 rows (535 parseable verdicts): judge-correct 83 / 83 / 86%

and mean quality 3.63 / 3.67 / 3.81 (of 4) for monolithic / agentic / swarm. The 3B judge is systematically lenient against the binary grader but preserves the ordering; it serves as corroboration, not as a result.

7 Threats to validity

Single model family; two accelerator classes but one vendor stack (Ollama) by design — stack effects are now a documented, guarded variable rather than an assumption. The agentic counterfactual is post-hoc (though rule-fixed before computation) and reported alongside, never instead of, the pinned metric. Swarm-1.0's inertness means the study so far bounds only *prompt* effects of the swarm topology; ensemble effects await SWARM 2.0. Cross-domain accuracy comparisons remain descriptive (bands sit at 42/58/67%). N=5 at temperature 0 measures serving jitter, not sampling variance.

8 Next steps

(1) HPC confirmatory sweep (A40, CUDA): running now, ETA within the day; data intake, integrity checks, the same counterfactual re-parse, judge pass, and analysis follow. (2) SWARM 2.0: temperature probe, amendment, then the corrected cell on HPC — run *in parallel with* (3) the Azure AI Foundry exploratory arm (thin OpenAI-compatible client, per-row cost accounting, N=1 cost pilot first). (4) Full paper draft from the assembled findings. (5) Stretch (time permitting, in priority order): a framework-as-agentic-layer A/B (LangChain/CrewAI vs. the hand-rolled loop — a direct extension of the false-revision finding), HPC scaling legs (32B, Qwen3.6-35B), and a budget-sensitivity leg at max_tokens 12288.

All raw rows, traces, amendments, the counterfactual script (`scripts/agentic_counterfactual.py`), and judge outputs are committed; every number in this report is reproducible from `results/` at the repository head.